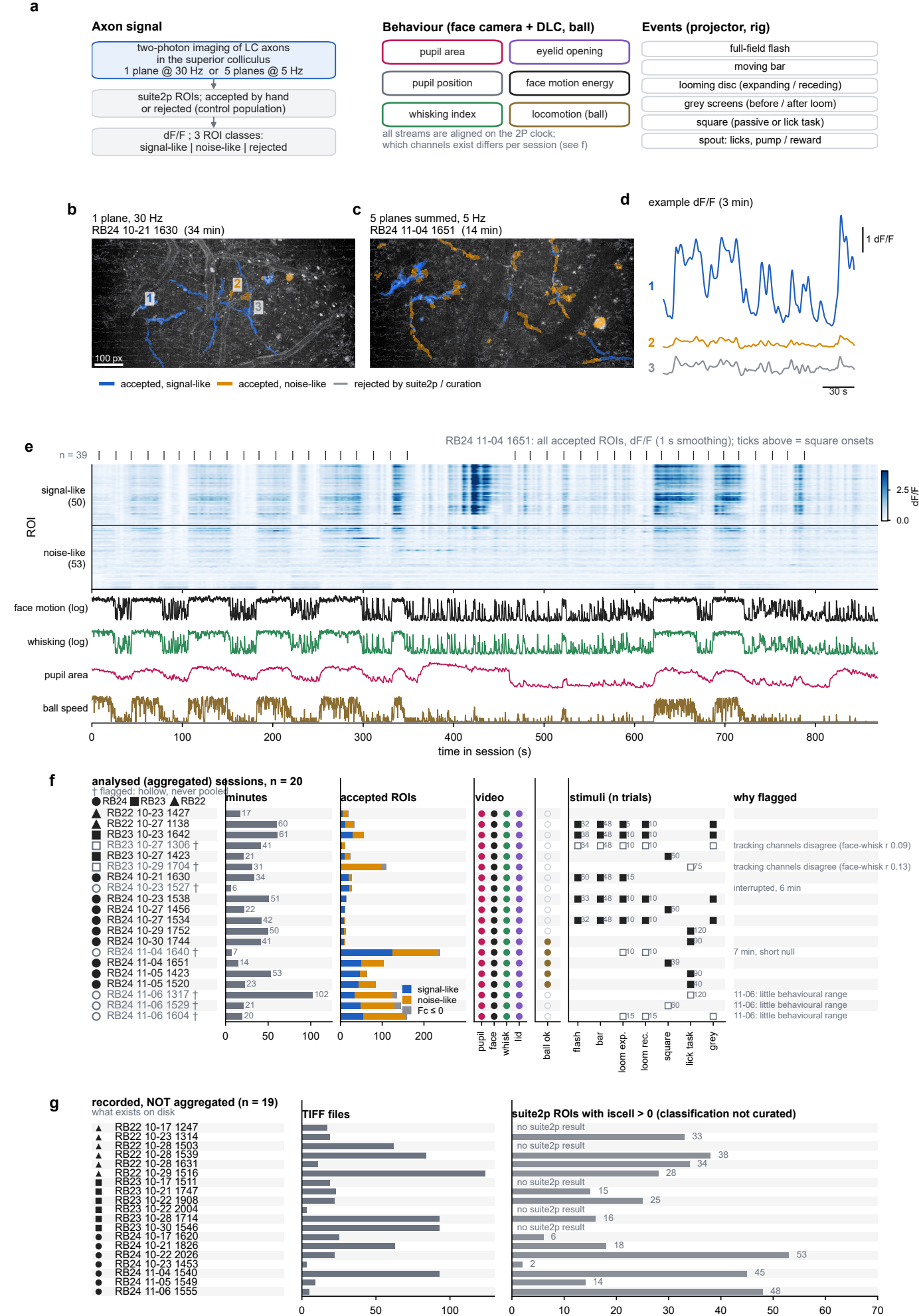

a
Axon signal
two-photon imaging of LC axons in the superior colliculus 1 plane @ 30 Hz or 5 planes @ 5 Hz
suite2p ROIs; accepted by hand or rejected (control population)
dF/F ; 3 ROI classes: signal-like | noise-like | rejected
Behaviour (face camera + DLC, ball)
pupil area
eyelid opening
pupil position
face motion energy
whisking index
locomotion (ball)
all streams are aligned on the 2P clock; which channels exist differs per session (see f)
Events (projector, rig)
full-field flash
moving bar
looming disc (expanding / receding)
grey screens (before / after loom)
square (passive or lick task)
spout: licks, pump / reward
b
1 plane, 30 Hz
RB24 10-21 1630 (34 min)
100 px
c
5 planes summed, 5 Hz
RB24 11-04 1651 (14 min)
accepted, signal-like
accepted, noise-like
rejected by suite2p / curation
d
example dF/F (3 min)
1 dF/F
30 s
e
RB24 11-04 1651: all accepted ROIs, dF/F (1 s smoothing); ticks above = square onsets
n = 39
ROI
signal-like (50)
noise-like (53)
2.5
dF/F
0.0
face motion (log)
whisking (log)
pupil area
ball speed
time in session (s)
f
analysed (aggregated) sessions, n = 20
† flagged: hollow, never pooled
RB24 RB23 RB22
minutes
accepted ROIs
video
stimuli (n trials)
why flagged
RB22 10-23 1427
RB22 10-27 1138
RB23 10-23 1642
RB23 10-27 1306 †
RB23 10-27 1423
RB23 10-29 1704 †
RB24 10-21 1630
RB24 10-23 1527 †
RB24 10-23 1538
RB24 10-27 1456
RB24 10-27 1534
RB24 10-29 1752
RB24 10-30 1744
RB24 11-04 1640 †
RB24 11-04 1651
RB24 11-05 1423
RB24 11-05 1520
RB24 11-06 1317 †
RB24 11-06 1529 †
RB24 11-06 1604 †
signal-like
noise-like
Fc ≤ 0
pupil
face
whisk
lid
ball ok
flash
bar
loom exp.
loom rec.
square
lick task
grey
tracking channels disagree (face-whisk r 0.09)
tracking channels disagree (face-whisk r 0.13)
interrupted, 6 min
7 min, short null
11-06: little behavioural range
11-06: little behavioural range
11-06: little behavioural range
g
recorded, NOT aggregated (n = 19)
what exists on disk
RB22 10-17 1247
RB22 10-23 1314
RB22 10-28 1503
RB22 10-28 1539
RB22 10-28 1631
RB22 10-29 1516
RB23 10-17 1511
RB23 10-21 1747
RB23 10-22 1908
RB23 10-22 2004
RB23 10-28 1714
RB23 10-30 1546
RB24 10-17 1620
RB24 10-21 1826
RB24 10-22 2026
RB24 10-23 1453
RB24 11-04 1540
RB24 11-05 1549
RB24 11-06 1555
TIFF files
suite2p ROIs with iscell > 0 (classification not curated)
no suite2p result